

Yurii Zaika

AI / Python Engineer

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Ten years of production engineering plus a personal track in Python and ML. Built a real-time speech pipeline: faster-whisper on CUDA, pyannote diarization, VAD and streaming — 55 modules, 15,500 lines of Python, with regression measurements of latency and RTF. Alongside it: data pipelines on SQLAlchemy and Playwright, and daily work with AI agents.

AT A GLANCE

- 10 years — in production retail systems, 2016 to now
- 15 500 lines — of Python in the speech pipeline: 55 modules, STT and diarization
- 27 — repositories in 2026: freelance and personal projects
- 6 sources — in the listings pipeline: OLX, LUN, Rieltor, Metrazh, Telegram, FB

WHAT I DELIVERED

- Leading the trading system client migration from Delphi 6 to Delphi 12/13, with a purpose-built tool that diffs forms and database mappings between versions so the switch does not break store operations.
- Architected and built the network's SMS notification platform: SMPP and GSM API, .NET Web API and MS SQL, designed for peak-volume campaigns.
- Built a real-time speech pipeline in Python: faster-whisper on GPU, pyannote diarization, VAD and streaming capture — 55 modules and 15,500 lines of code.
- Wrote its quality regression suite — RTF, latency, noise robustness, accuracy ceilings — and drive decisions from the numbers rather than impressions.
- Built Python data pipelines: SQLAlchemy 2 and Alembic over PostgreSQL, Playwright for protected pages, six listing sources merged into one catalog.

STACK

Data: MS SQL Server, T-SQL, PostgreSQL

Quality and delivery: xUnit / NUnit, NSubstitute / Moq, AutoFixture, FluentAssertions, SonarQube, Jenkins, Docker, Serilog / NLog, Elastic APM, Git / приватний NuGet

Engineering practice: Legacy modernization, Modular monolith, Three-tier architecture, Integration design, Production ownership

Python: Python 3.11, FastAPI, SQLAlchemy 2, Alembic, Pydantic, Typer CLI, Playwright, selectolax, NumPy, pytest, ruff / black, Jinja2

Speech and ML: faster-whisper (CUDA), pyannote.audio, VAD, Speaker diarization, Quality evaluation: RTF, latency, Streaming audio pipelines, WASAPI / sounddevice, PyTorch / GPU

Working with AI agents: Claude Code, MCP-сервери, Custom subagents and hooks, Agent tooling (CDP), Prompt engineering

JS / TS ecosystem: Next.js 16, React 19, NestJS, Prisma, Node.js

KEY PROJECTS

Real-time speech pipeline (ORK) (2026) — Author

Real-time speech recognition from calls: parallel microphone and system-audio capture, VAD-based segmentation, faster-whisper on GPU, pyannote diarization with incremental clustering, translation. A separate regression suite — RTF, latency under load, noise robustness — shows exactly where quality is lost.

Python 3.11, faster-whisper, pyannote.audio, CUDA, FastAPI, NumPy, WASAPI

Real-estate data pipeline (2026) — Author

A listings parser with pagination, normalization and price history: SQLAlchemy 2 models with Alembic migrations over PostgreSQL, two transports — plain HTTP and Playwright for protected pages — a Typer CLI, pydantic-settings config and pytest coverage.

Python, SQLAlchemy 2, Alembic, PostgreSQL, Playwright, selectolax, Typer, pytest

Legacy system map (2026) — Author

A tool for mapping the trading system before migration: Python parses .dfm and .pas files, extracts stored-procedure calls and form relations, and a Next.js app renders it as an annotated map.

Python, Next.js, TypeScript, PostgreSQL

AM Mobility — web and mobile app (2026) — Full stack developer

Client project: a NestJS back end with Swagger and PostgreSQL, a Next.js web app, and an Expo / React Native mobile app with MapLibre maps, geolocation and offline state.

NestJS, Next.js, Expo, React Native, PostgreSQL, MapLibre, TanStack Query

Trading client migration: Delphi 6 to Delphi 12 (2024 — 2026) — Migration lead

Moving the trading system client (955 units, 828 forms) to a modern compiler with a core redesign. Built a separate tool that diffs forms and table mappings between the old and new versions, so the migration runs without stopping stores.

Delphi 12/13, Object Pascal, VCL, MS SQL, DevExpress, FastReport

https://github.com/qwert11/d6_d13_dbmapping_checker

SMS Messenger (2020 — 2023) — Architect and developer

The network's SMS notification platform: an in-house gateway to carriers over SMPP and to modems over GSM API, campaign queues, an API for internal systems.

ASP.NET Core, Web API, SMPP, GSM API, MS SQL, Hangfire

EXPERIENCE

Freelance and personal projects — AI / Python and Full Stack Developer

2026-03 — present · Dnipro · remote

- A real-time speech pipeline: faster-whisper on CUDA, pyannote diarization, VAD, WASAPI capture — 55 modules, 15,500 lines of Python.
- And a regression suite for it: RTF, latency under load, noise robustness, quality ceilings — decisions come from measurements, not impressions.
- A real-estate parser in Python: SQLAlchemy 2 and Alembic over PostgreSQL, Playwright and selectolax, a Typer CLI, pytest and ruff.
- A listings pipeline over six sources with geocoding, deduplication and catalog publishing through GitHub Actions.
- A legacy-code analyzer: Python parses .dfm and .pas files and diffs stored-procedure calls, Next.js renders the system map with annotations.
- Client work: NestJS with Prisma and PostgreSQL, Next.js 16 with React 19, a mobile app on Expo with MapLibre maps.
- Daily work with AI agents: MCP servers, custom subagents and hooks, and my own CDP browser inspector as agent tooling.

ATB-Market — Lead Application Programmer

2016-06 — present · Dnipro

- Own the client side of the network's trading system: 955 units, 828 forms, MS SQL, covering supply documents, pricing and fiscal invoices.
- Since 2018 also build internal services on ASP.NET Core and Angular: access, audit, reporting, integrations — 33 of the network's systems carry my code.
- UAM: access management for 2,000+ users on OpenID Connect, ADFS and Kerberos.
- Network integrations: EDI and e-document flow, M.E.Doc, Oracle EBS, 1C, Kafka, BizTalk, SSIS and SSRS.
- Fiscal-law changes delivered end to end: excise stamps and AED, UKTZED, customs simplifications, VAT rates, in a live retail network.
- Leading the client migration from Delphi 6 to Delphi 12/13 with an in-house form and DB-mapping diff tool.
- Engineering practice: xUnit and NSubstitute tests, SonarQube and Jenkins, Serilog, Docker, a private NuGet feed.

EDUCATION AND CERTIFICATES

- Computer Engineering · computer systems and networks — Zaporizhzhia Institute of Economics and Information Technologies (2008 — 2011)
- Mining Engineer, underground development — Kryvyi Rih Technical University (1998 — 2003)
- Angular 9 — Luxoft (2021)
- Programming in Delphi — Institute of Modern Professions (2006)

CODE ON GITHUB

- Housing catalog across nine regions — six-source aggregator with a map, price indices and a GitHub Actions build
<https://github.com/qwert11/budynky-zahid>
- ML Formula Wiki — 72 machine-learning formulas from Raschka's book, rendered with KaTeX

- <https://github.com/qwert11/ML-Formula-Wiki>
- M3 to M4 Mapping Checker — diffs forms and stored-procedure calls between Delphi 6 and Delphi 13
https://github.com/qwert11/d6_d13_dbmapping_checker
- TeamBeacon — a real-time beacon for finding your group in a crowd: ASP.NET Core, SignalR, Docker
<https://github.com/qwert11/TeamBeacon>
- SQL Profiler Normalizer — turns profiler sp_executesql output into a readable EXEC, fully client-side
<https://github.com/qwert11/sql-profiler-normalizer>
- Angle Tool — measures line angles on charts and screenshots in the browser
<https://github.com/qwert11/slope-angle>
- Resume Engine — this site: YAML to Jinja to HTML, PDF, DOCX and Markdown in two languages
<https://github.com/qwert11/resume2>

LANGUAGES

- Ukrainian — native
- English — technical documentation and written communication; spoken at basic level